NEEV ZERONI

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EXPERIENCE

ABBOTT, Plymouth, Minnesota || **R&D Engineering Co-Op**, 2025-Present

- Co-lead cadaver planning, study, and analysis to characterize offset from in vitro to in vivo testing.
- Lead device baseline and coaptation study and analysis, referencing competitive device benchmarking and literature.
- Conducted 14 simulated device use tests with respective analysis.
- Lead unique device characterization method development using micro-CT analysis, with over 60 devices analyzed.
- Designed in CAD and fabricated fixtures and method improvements on fatigue testers for test method development.

VENTUREMED GROUP, Plymouth, Minnesota || **R&D and Ops Engineering Intern**, 2023, 2024

- Developed and built new fixtures in CAD to improve production efficiency and reduce cost, ensuring compliance with regulatory and quality requirements, reducing shipment costs by 30%, and enabling production scale up.
- Led sample preparation for sterilization, including sterilization data review and documentation.
- Optimized vision system at incoming inspection, leading to improved quality and visual communication.
- Reverse-engineered medical devices to generate sub-assemblies, enabling a faster prototype lead time at lower cost, saving $10K and 3 months.

CROSSFIRE MEDICAL, Maple Grove, Minnesota || **Engineering Intern**, 2023

- Conducted root cause investigation and collaborated on device design changes to improve durability.
- Led post-study device review and documentation to gain FDA approval.
- Set up and calibrated equipment in the R&D lab to aid in design and research.
- Assisted in person during preclinical animal study.

VIGILANCE SAFETY, Plymouth, Minnesota || **Co-founder and Technical Lead**, 2022-Present

- Led design and technical development of an RFID tag and scanner system to mitigate school shootings and improve school safety.
- Developed new technology while managing ambiguity, self-learning, and collaborating with industry leaders.
- Streamlined contract manufacturers' engagements.
- Created, coded, and tested a functioning prototype. Tested and analyzed operational feasibility.
- Led supply chain and manufacturing operations development.
- Led community engagement and fundraising events.

NORTHSTAR ROBOTICS, Minneapolis, Minnesota || **Turret Lead**, 2024-2025

- Lead design, development, and testing of a Robomaster turret using CAD, DFM, and electronic integration.
- Designed custom ball path, motor mounts, and assembly configurations.
- Used CNC and manual machining equipment to fabricate parts.
- Conducted durability, jam, and fatigue tests to improve reliability.

FIRST ROBOTICS, Plymouth, Minnesota || **Mechanical Captain**, 2020-2024

- Served as team lead for mechanical and technical development. Provided hands-on teaching and mentoring to underclassmen.
- Experience with prototyping and failure mode analysis.
- Operated complex equipment: vertical bandsaw, 3D printing, milling, metal, and woodworking tools.
- Developed game strategy to optimize competitive design.

Projects

Vortec 5.3 V8 Rebuild and Project Car

- Installed new cams, oil pump, intake, and gaskets.
- Dropped into a 2ng gen Ford Ranger with a new transmission, rear axle, and fuel system.

RI3D (robot in 3 days) Robot

- Created and documented a FIRST robot in 3 days.
- Developed model in Solidworks CAD and used manual and CNC machining processes to create the robot itself.

Robotic Quadruped

- Fully developed in CAD, optimized with FEA. AI integration coming soon.

Custom FFB Sim Wheel

- Designed using scrap motors and custom PCBs.

EDUCATION

UNIVERSITY OF MINNESOTA, Minneapolis, Minnesota || **Sophomore, Mechanical Engineering**